MIL-H-20638B

31 March 1969
SUPERSEDING
MIL-H-20638A
8 June 1964

MILITARY SPECIFICATION

HOOK, CANT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of cant hook.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-I-666 - Iron Castings, Malleable.
QQ-S-631 - Steel Bar, Carbon, Hot Rolled (Special Quality).

MILITARY

MIL-H-15424 - Hand Tools, Packaging Of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

FSC 5120

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* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2016-65 - Standard Methods of Test for Moisture Content of Wood.

E 18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a sample of the cant hook covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standard product.- The cant hooks delivered under this specification shall be the manufacturer's current, standard commercial product, except for any changes necessary to comply with specification requirements.

3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for cant hooks, provided the completed item complies with all provisions of this specification.

* 3.3.1 Malleable iron.- Malleable iron shall conform to grade 1 of QQ-I-666.

* 3.3.2 Steel.- Bar steel shall conform to Steel Numbers 1060 to 1090 of QQ-S-631.

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* 3.3.3 Wood.- Wood shall be hickory, with not more than 22 growth rings per inch, or rock maple, with not more than 12 growth rings per inch. Hickory shall weigh not less than 45 pounds and maple not less than 44 pounds per cubic foot at 12 percent moisture content (see 4.4.2).

3.4 Design and construction.- The cant hook shall consist of an extension toe ring, a cant hook clasp, a bolt and nut, a hook and a handle. All components shall be replaceable.

3.4.1 Extension toe ring and cant hook clasp.- The extension toe ring and clasp shall be fabricated of malleable iron specified in 3.3.1 or forged steel, and shall fit tight on the handle. The extension toe ring shall be the multi-point bite type.

* 3.4.2 Hook.- The hook shall be a commercial duck bill type, not less than 1 inch wide by not less than 7/16 inch thick, and between 8 and 9 inches long, measured from the point to center of hole, forged from steel bar specified in 3.3.2, and hardened to Rockwell B 96 - 100 (see 4.4.1).

* 3.4.3 Bolt and nut.- The bolt shall be the manufacturer's standard, steel, square headed bolt, complete with compatible square or hex nut. When the nut is fully tightened on the assembled cant hook, the bolt shall not extend more than 1/4 inch beyond the nut.

3.4.4 Handle.- The handle blanks shall be fabricated of wood specified in 3.3.3. The wood blanks shall have a moisture content not exceeding 15 percent at the time of fabrication of handle (see 4.3.2.1). The slope of grain of the handle blank shall not exceed 1 inch in 10 inches when tested in accordance with 4.4.3. The overall length shall be 54 inches plus or minus 1 inch. The large diameter at clasp point shall be not less than 2-1/2 inches and shall taper down at one end to receive the formed toe ring and at the opposite end to a minimum of 1-1/2 inches below the end knob.

3.5 Finish.- All metal parts shall have a black oxide coating. The wood handle shall be smooth and finished in accordance with the manufacturer's standard commercial practice.

3.6 Marking.- The cant hook shall be marked with the manufacturer's identification or readily identifiable trademark.

3.7 Workmanship.- All exposed surfaces shall be smooth and free of sharp edges (except toe ring and hook points), and shall be free of rust, scale, burrs, fins, nodules, cracks, blowholes and plugging.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in 4.3.3.1, dimensions specified, and tested as specified in 4.4.1.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Intermediate testing.-

4.3.2.1 Testing for moisture content of wood blanks.- Moisture content of the wood blanks requirement in 4.4 shall be determined in accordance with ASTM D 2016-62 at time of fabrication, except that when testing is performed by the electric moisture meter method, three determinations shall be made on each sample unit. The average of the three determinations shall be the moisture content of the sample unit. The lot shall be all handle blanks prepared under the same conditions, stored as a group, and offered for inspection at one time. The sample unit shall be one handle blank of wood ready for turning. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

4.3.2.2 Hardness.- The hook shall be tested for hardness in accordance with 4.4.1 to determine compliance with 3.4.2. A lot shall consist of all hooks offered for inspection at one time. The sample unit shall be one hook. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

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4.3.2.3 Slope of wood grain.- The slope of grain of the wood blanks specified in 3.4.4 shall be determined in accordance with 4.4.3. The sample lot shall consist of all wood handle blanks offered for inspection at one time. The sample unit shall be one wood blank. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

4.3.3 End item inspection.- For sampling purposes, the lot shall be expressed as the number of cant hooks offered for inspection at one time. The sample unit for the inspection shall be one cant hook.

4.3.3.1 Visual examination.- The end item shall be examined for defects in table I. The inspection level shall be level II with an AQL of 2.5, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect
Finish (metal components)	Not smooth, not free from scale, rust, burrs, fins, nodules, cracks, blowholes or plugging Not black oxide coated
Handle	Not finished as specified
Design and construction	Hook not duck bill type Extension toe ring not two point type Clasp not securely attached Handle not securely set in extension toe Hook not attached to clasp adequately; too tight; too loose; wrong bolt or nut Bolt extends more than 1/4 inch beyond nut
Marking	Missing, illegible, incomplete

4.3.3.2 Dimensional examination.- Inspection shall be made of cant hooks for compliance with dimensions specified herein. Any dimension not within the specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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4.3.4 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with MIL-H-15424.

4.4 Tests.-

4.4.1 Test for hardness of metal.- The hardness tests shall be conducted in accordance with ASTM E 18-65.

* 4.4.2 Weight of wood.- The weight per cubic foot of the wood for handles specified in 3.3.3 shall be determined by obtaining moisture content in accordance with ASTM D 2016-65 and then calculating the weight per cu. ft. from the measurements of the samples used therein.

* 4.4.3 Slope of wood grain.- Slope of grain may be determined by combining the grain direction of two adjacent faces of a wood handle blank (see 3.4.4). When one of these faces is straight grained, the true slope of grain is shown on the other face. When there is a slope on both faces, the true or combined slope is greater than the slope on either face. Slope of grain is shown on edge grained faces by the summer wood bands, by the direction in which a free flowing ink spreads or by the course taken by a narrow strip lifted by a knife point and torn out. Direction of fiber may also be determined by the Teco slope grain detector as supplied by the Timber Engineering Co., Inc., Washington, D. C.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with MIL-H-15424.

6. NOTES

6.1 Intended use.- The cant hook covered by this specification is intended to provide leverage when turning poles, logs and similar items.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When preproduction sample is required (see 3.1).
(c) Selection of applicable levels of preservation, packaging and packing (see 5.1).

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6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, corrections, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Navy - YD
Air Force - 84, 85

User activities:

Army - CE, ME
Navy - MC

Preparing activity:

Army - GL

Project No. 5120-0573

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Hook, Cant MIL-H-20638B

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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